



Speech stimulus design to study the neural coding of speech and the impact of cochlear synaptopathy

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Abstract

Since auditory neural coding mechanisms have been uncovered using simple artificial stimuli, the nature of their involvement in speech perception has remained unclear. However, using analysis-resynthesis techniques, it is possible to design naturalistic speech stimuli that present the same appeal as artificial tones in terms of precise control and parametrisation. Here, speech stimuli were designed to assess the involvement of a specific coding mechanism: the coding of temporal fine structure through phase-locking. This mechanism is supposed to be impaired in cochlear synaptopathy, a form of hearing loss that remains undetected through pure tone audiometry. The design process described here seeks to accommodate all the constraints that can arise from large multi-centric studies covering different species, methodologies, and languages.

Index Terms: speech analysis-resynthesis, hearing loss, neural coding

1. Introduction

Speech perception relies on the decoding of acoustic information into phonetic categories that are then strung into words, themselves sequenced into sentences [1]. This process has been tied to specific neural mechanisms that are responsible for encoding the temporal fine structure (TFS) of auditory stimuli [2]. TFS encoding is thought to enable precise coding of frequency information through the phase-locked firing of the auditory neurons. Consequently, TFS coding is limited by the phase-locking abilities of the auditory nerve, and only happens at frequencies low enough for synchronised firing to occur [3]. TFS coding abilities are hindered in case of cochlear synaptopathy (CS), a condition that reduces the number of neural fibres that can fire synchronously. CS is largely prevalent in older adults and is also known as “hidden hearing-loss” since it does not affect pure tone audiometry, which is the standard clinical diagnostic tool [4, 5]. It is believed to result in a degradation of speech intelligibility in noisy environments [6]. However, its hidden nature makes it difficult to diagnose in humans, and although the consequences of CS on the general coding of sounds has been documented in animals, the direct link between the neural disorder and the speech-in-noise perception deficit remains elusive.

To better understand the interplay between CS, the neural coding of TFS, and speech perception, a unified approach across humans and animals models need to be adopted. Behavioural accounts of speech perception can be obtained from human participants, as well as compound electrophysiological data (EEG, such as auditory brainstem responses, ABR) [7]. Animal models allow the same compound electrophysiological recordings, but also allow single unit recordings that offer a direct view of the degree of neural synchronisation with the TFS

(phase-locking) that the observed neuron is able to provide [8]. Computational modelling can then be used to combine the different approaches across the different species [9, 10]. However, for these data to remain comparable, it is crucial that the same stimuli are used in all the phases of such a project. Furthermore, using a simple discrimination task facilitates modelling and links to physiological recordings. Here, we describe how such stimuli were designed and produced, the constraints they had to satisfy, and how they could be generated in a flexible way.

2. Constraints

The design constraints are separated into two categories: *intrinsic requirements* that are directly induced by the purpose underlying the creation of the stimuli; implementation, or *operational requirements* that are imposed by the specific tasks and methods that will be used.

2.1. Intrinsic requirements

2.1.1. Ecological validity

To be generalisable to speech perception at large, the stimuli have to capture some of the essential traits of speech. To satisfy this requirement it seems obvious that (a) the stimuli must consist of, or be as close as possible to, real speech excerpts. This also means the speech stimuli must (b) pertain to the language of the (human) listeners involved in the experiments.

2.1.2. Sensitivity to cochlear synaptopathy

Because we are interested in the effects of CS, we want to focus on speech features that are potentially sensitive to variation in TFS processing abilities. In other words, we are interested in speech contrasts that either do or do not involve TFS processing. It is assumed that TFS coding requires neural phase-locking, which is limited in frequency. There is much debate about what the upper frequency limit of phase-locking may be [3], with proposed values between 1.5 and 10 kHz. Based on this review, it can be speculated that TFS would be optimally coded below 1.5 kHz, but that coding would progressively deteriorate above this frequency, and eventually entirely vanish above 10 kHz. Therefore, for our current purpose, we have chosen a TFS processing limit of 1.5 kHz, meaning that frequencies below this limit may be coded using TFS mechanisms, while frequencies above this limit are not guaranteed to be coded this way. Note that the frequency importance function for speech information peaks around 1.5 kHz [11], indicating that frequencies on both sides of this limit are likely to contribute comparably to intelligibility.

2.1.3. Steady-state and transient

Lorenzi *et al.* [2] linked TFS perception abilities to speech perception using an identification task for stimuli in the form /aCa/ where C is a consonant. Unlike vowels that tend to be steady-state, consonants are transient sounds where the spectral content is changing rapidly. This raises the possibility that TFS might be specifically involved in the perception of transients. However, neural responses to non-stationary sounds are more difficult to analyse than those to steady-state sounds for which long-term spectra can be more easily interpreted. As a result, it is desirable to create both steady-state and transient stimuli to cover both aspects.

2.2. Operational requirements

2.2.1. Duration of stimuli

Natural speech mostly consists of rather long utterances like sentences that have a duration in the order of seconds. However, ABR measurements require a large number of repetitions [12]. For this reason, it is usually performed with very short stimuli, < 100 ms. Similarly, single unit recordings – which record the activity of single auditory neurons – are performed on live, anaesthetised animals using a procedure that is time bound [8]. Therefore, this method also benefits from using short stimuli. Regarding the behavioural task in humans, it would also require a fair number of repetitions to be relatable to other measurements. In addition, the test is likely to be embedded into a much larger test battery in order to provide complete audiological assessment of the participant. As a result, it is also preferable to keep the behavioural test as short as possible, which also points towards shorter stimuli.

2.2.2. Continuum of stimuli

The stimuli are to be organised in pairs contrasting specific phonemic contrasts. Behavioural tasks could then simply be identification or categorisation of the phoneme. However, to ensure that the task difficulty can be adjusted, it seems desirable to have a continuum of stimuli ranging from one end of the contrast to the other. This will allow pre-testing to select pairs of items that have a pre-determined level of behavioural discriminability. Furthermore, in studies involving participants with impaired hearing, it is common to be faced with the issue that while the normally hearing group is at ceiling, the group with hearing loss is at floor performance. To keep comparisons meaningful and interpretable, it is recommended to use adapted levels of difficulty for the two groups. Continua of stimuli will make it possible to select pairs of items that are suitable for listeners with various listening abilities without the concern of being stuck at floor or ceiling performance.

Regarding the physiological recordings, it is also desirable to obtain continua of stimuli. Indeed, when comparing only two stimuli in a stochastic process like neural activity, it is difficult to reliably relate observed differences in activity to the stimuli themselves. With a continuum, more powerful regression analyses can be used to relate the continuous change in neural activity to the continuous change in acoustic information in the stimulus.

2.2.3. Controlled f_0 cycles

In speech, vowels are voiced sounds, meaning that they have a fundamental frequency (f_0). Although the stimuli are meant to be as natural as possible (2.1.1), it is also desirable that they

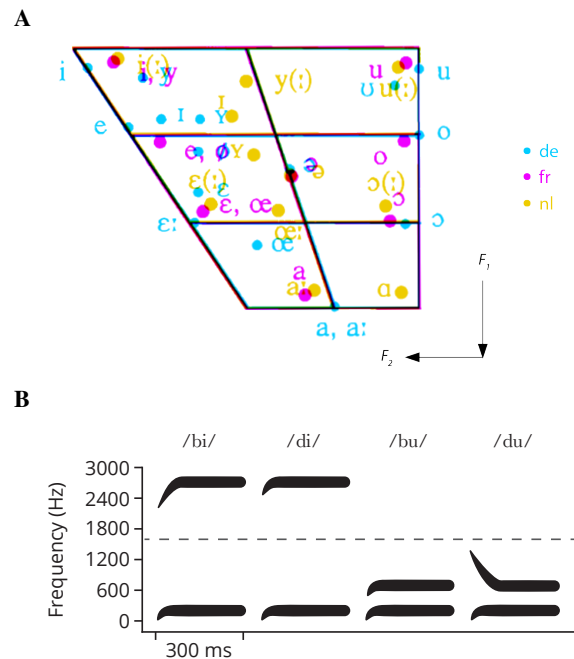


Figure 1: **(A)** Vocalic triangle for German ('de', cyan), French ('fr', magenta) and Dutch ('nl', yellow). This shows the vowel inventories for the three languages by placing each vowel (represented by their IPA symbol) in their estimated relative position in terms of first (F_1) and second (F_2) formants. Adapted from [13]. **(B)** Schematic spectrograms showing F_1 and F_2 trajectories for /bi/, /di/, /bu/ and /du/. The dashed line shows the 1.5 kHz TFS boundary. Adapted from Delattre et al. [14].

have a known, fixed f_0 . Since f_0 determines the periodicity of the waveform of the stimuli, their neural response is synchronised to the f_0 . In the analysis of ABRs, spectral analyses through FFT are designed to be aligned with the f_0 to maximise the accuracy. As a result, stimuli with constant f_0 produce stronger apparent responses.

3. Implementation

3.1. Phonemic contrasts

Two constraints apply to the choice of phonemic contrasts: 2.1.1.b implies that the considered phonemes have to belong to the inventory of the language of the human participants considered for this material. In our case, participants could be German, Dutch or French. We thus looked for phonemic contrasts that were common to these three languages, and that also satisfied 2.1.2 and 2.1.3.

To satisfy 2.1.2, we selected phonemic contrasts that are generally realised through acoustic traits that are either below or above the chosen 1.5 kHz limit. To satisfy 2.2.1 (keep stimuli short), we selected single vowels for the steady-state stimuli, and consonant-vowel (CV) syllables for the transient stimuli.

3.1.1. Steady-state stimuli: vowels

We reviewed the vowel inventories of German, French and Dutch as listed by the International Phonetic Association [13]. First, when superimposing the canonical vocalic triangles, it appears that there is some overlap between languages (Figure 1A).

Table 1: Selected vowel and syllable contrasts followed by example words in German, French and Dutch for illustration purposes.

			German	French	Dutch
Vowels	< 1.5 kHz	/o/	oder (<i>or</i>)	ôter (<i>to remove</i>)	oké (<i>okay</i>)
		/u/	Universität (<i>university</i>)	outil (<i>tool</i>)	oester (<i>oyster</i>)
	> 1.5 kHz	/i/	ikonisch (<i>iconic</i>)	ignorant (<i>ignorant</i>)	iets (<i>something</i>)
		/y/	über (<i>over</i>)	université (<i>university</i>)	universiteit (<i>university</i>)
Syllables	< 1.5 kHz	/du/	Du (<i>you</i>)	doux (<i>soft</i>)	doe (<i>do</i>)
		/bu/	Butan (<i>butane</i>)	bout (<i>end</i>)	boete (<i>fine</i>)
	> 1.5 kHz	/di/	die (<i>the</i>)	dit (<i>said</i>)	die (<i>that</i>)
		/bi/	Bier (<i>beer</i>)	bipède (<i>biped</i>)	bier (<i>beer</i>)

Further collecting actual formant values from various studies [15, 16, 17, 18], they can be projected onto the same (F_1 ; F_2) plane. We selected the /o/-/u/ contrast that concerns only F_1 in a region far below the putative TFS limit of 1.5 kHz, i.e. this is a contrast that could rely on TFS coding. As contrast that is meant not to depend on TFS coding, we selected /i/-/y/, where the F_2 difference is located above 1.5 kHz. Example words in the three considered languages are shown in Table 1.

3.1.2. Transient stimuli: syllables

Although Lorenzi *et al.* [2] used speech tokens of the form vowel-consonant-vowel, ABR recordings have been popularized with shorter CV syllables [12] which we also adopted here to satisfy 2.2.1.

Again, examining the consonant inventories of German, French and Dutch, there is considerable overlap. Delattre *et al.* [14] described schematic transients for CV syllables starting with /b/, /d/ and /g/. Since /g/ is not present in Dutch (as the letter ‘g’ is realised as /x/), we focused on /b/ and /d/. We selected two vowels, /i/ and /u/ that are involved in the vowel contrasts, and that provide syllable contrasts above and below the 1.5 kHz putative TFS limit as illustrated in Figure 1B. As a result, /du/-/bu/ is a contrast that happens below the 1.5 kHz boundary and may involve TFS coding; whereas /di/-/bi/ is a contrast that happens above the 1.5 kHz limit and therefore does not involve TFS coding. Again, example words are reported in Table 1.

3.2. Analysis-resynthesis from voice recordings

To ensure that the stimuli are natural and capture realistic traits of speech (2.1.1), we used actual recordings from a male speaker rather than relying solely on speech synthesizers. The recordings were then analysed, i.e. decomposed, using the WORLD vocoder [19, 20, 21].

The recordings were obtained using an SM58 microphone (Shure, Niles, Illinois, USA) and an EVO4 soundcard (Audient, Herriard, Hampshire, UK). A sampling rate of 44.1 kHz and a 32-bit (float) precision were used. The talker was a 43-year-old, French native speaker measuring 1.85 m, with no history of speech production disorder and no history of smoking. The recordings had an average f_0 of about 105 Hz.

To decompose the recordings using WORLD, the f_0 was first extracted using the *dio* and *stonemask* methods [22]. The spectral envelope was then estimated using *cheaptrick* [23]. Finally, the spectro-temporal aperiodicity map was estimated using *D4C* [19].

For all the stimuli, a spectral envelope was carefully prepared according to the constraints listed above. The stimuli were then be resynthesised with a specified f_0 and duration.

3.2.1. Vowels

The average spectral envelope of each vowel was approximated with a sum of Gaussian functions. For /o/-/u/, all the Gaussian components above 1.5 kHz were aligned, to keep differences only < 1.5 kHz. The continuum between the two vowels was realised by shifting a single Gaussian component from 200 Hz (/u/) to 380 Hz (/o/). For /i/-/y/, the components below 1.5 kHz were aligned, while the ones above were shifted from 1.70, 2.10, 3.00 kHz (/y/) to 1.95, 2.80, 4.00 kHz (/i/), respectively. The produced vowels are displayed in panels (A) and (B) of Figure 2.

3.2.2. Syllables

For the syllables, no Gaussian mixture was used because the /d/-/b/ contrasts do not solely rely on differences in formant trajectories. Instead, spectral-temporal regions were defined in the spectral envelope matrix, such that no difference between stimuli could occur after 100 ms. For /du/-/bu/, the region comprised before 100 ms and below 1.5 kHz contained either /du/ or /bu/, whereas the region above 1.5 kHz contained the average of the spectral envelopes of /du/ and /bu/. The segment after 100 ms contained the spectral envelope of the /u/-sound described above. For /di/-/bi/, the procedure was the same except that changes only permitted above 1.5 kHz, and the segment after 100 ms contained the spectral envelope of the /i/-sound described above. For all the stimuli, great care was dedicated to aligning the onsets of the high and low frequency parts. Continuum steps were achieved by using weighted sum of the spectral envelopes of the items in a pair in the region allowing change. The produced syllables are displayed in Figure 3.

3.2.3. Post-processing

Early versions of the stimuli produced unexpected responses in animal recordings, showing a strong high frequency component (> 10 kHz). While this component would be unlikely to be heard in humans, most rodents used in animal studies have a hearing range that is shifted towards the high frequencies relative to humans. To avoid unwanted effect on physiological recordings, the component had to be controlled. A detailed examination of the stimuli revealed that, for some of the items, a step was introduced between two samples, right before each glottal pulse. It is suspected that this small artefact may be generated in the PSOLA resynthesis procedure. To neutralize the artefact, all stimuli were low-pass filtered below 9 kHz with a 4th-order zero-phase Butterworth filter.

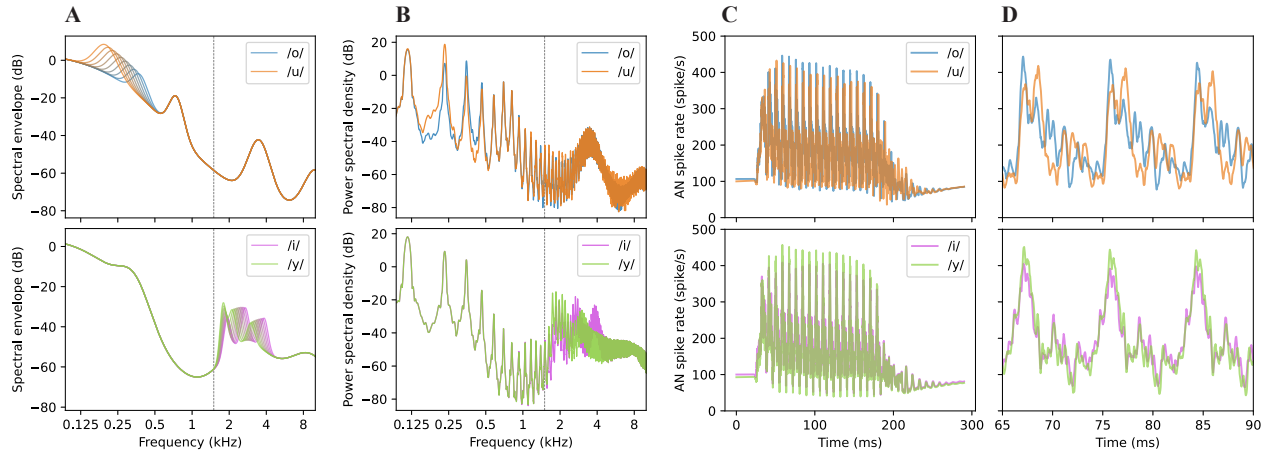


Figure 2: Vowel stimuli. (A) Spectral envelopes of the /o/-/u/ (top) and /i/-/y/ (bottom) continua. (B) Spectra of the resynthesised endpoints of the continua. (C) Simulated compound action potential in the auditory nerve in response to the endpoints of the continua (see text). (D) Same but zoomed in on a detail, illustrating how the TFS in response to /o/ differs from that of /u/ (top), while there is no such difference between /i/ and /y/.

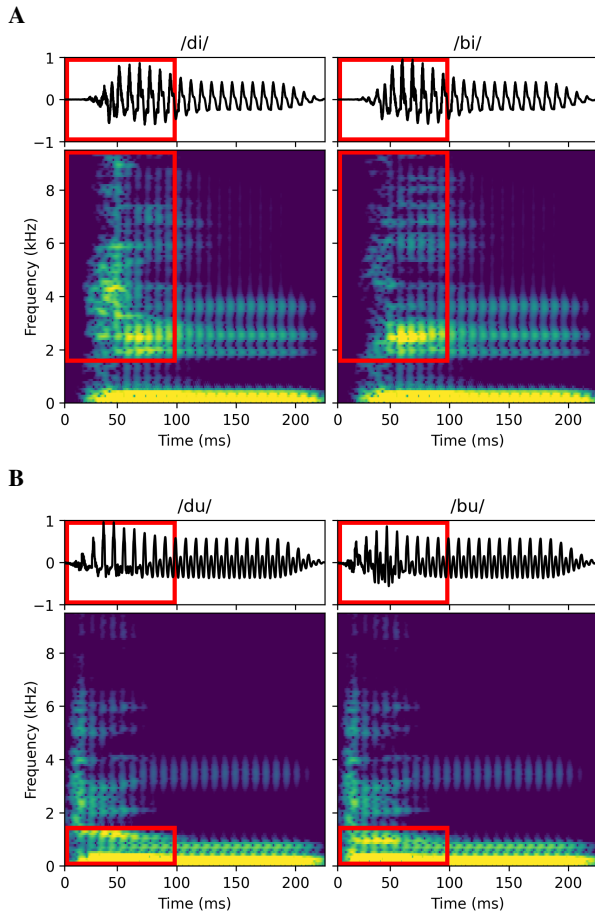


Figure 3: Syllable stimuli. (A) Waveform and spectrum for the /di/ and /bi/. The red rectangle highlights the spectro-temporal region the differences between the syllables were confined to. (B) Same for /du/ and /bu/.

4. Evaluation

To test whether the generated stimuli can indeed elicit the type of neural response that is expected, we used a computational model of human auditory system (UR_AN_IC_Model_2024c) [24, 25]. The responses to our stimuli presented at a level of 70 dB SPL are displayed in panels (C) and (D) of Figure 2. The details show that, as expected, the phase-locked neural response to /o/ differs from that of /u/ (D, top). In contrast, the phased-locked neural response to /i/ does not differ markedly from that of /y/. In other words, the model indicates that designed stimuli are fit for their purpose: one pair of vowels does elicit differences in TFS that rely on neural phase-locking, while the other pair of vowels solely relies on place coding.

Applying the model to the syllables shows similar differences, albeit restricted to the transient period of the stimuli.

5. Conclusions

The method described here, and the preliminary results obtained from modelling, indicate that it is possible to create natural speech stimuli that can be used in simple discrimination tasks while targeting specific neural coding mechanisms. A computational model of the human auditory system has been used to confirm that, as per design, the contrast between some of the generated stimuli is coded through the targeted neural coding mechanism (TFS), while others are not. The produced set of stimuli has since been used in neurophysiological experiments in animals, as well as in electrophysiological and behavioural experiments in humans [26, 27]. They are available at <https://doi.org/10.5281/zenodo.10322257>.

Overall, the followed approach demonstrates that current analysis-resynthesis techniques make it feasible to generate tightly controlled speech stimuli that still retain naturalness, while being designed for very specific purposes. The technique could be further extended to explore other neural coding mechanisms, and to assess various components of language more systematically.

6. Acknowledgements

This work was supported by ERA-NET NEURON CoSySpeech (FWO G0H6420N); VICI Grant (Grant No. 918–17–603) of the Netherlands Organization for Scientific Research (NWO) and the Netherlands Organization for Health Research and Development (ZonMw). The study was conducted in the framework of the LabEx CeLyA (ANR-10-LABX-0060/ANR-11-IDEX-0007) operated by the French ANR.

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